

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** MODULAR VALVES *****			814	NIPPLE TO BE CUT FROM SS SCH 160 PE PIPE.
20										B1	ALL BOLTS ARE METRIC DIA.
20			MODULAR DBB	K03AS02	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1
30		VAV	MODULAR DBB	K03AS02	3	3	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0		INCH 1500# PERAR BALL VALVES ITEM CODE
40			MODULAR DBB	M03AS02	0.5	2	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0		B15AM10
50		VAW	MODULAR DBB	M03ES02	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF SS A182 F316SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	3	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1
60											INCH 1500# PERAR BALL VALVES ITEM CODE
60		VB2					***** BALL VALVES *****			B15AS22	
70										FA8	USE THREADED FLANGES FOR INSTRUMENT
70			BALL	A03AS01	2	2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT RB LO	0	0		CONNECTIONS ONLY.
80			BALL	A03AS01	3	4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT RB LO	0	0	FF5	FLAT BOTTOM NIPOFLANGES FOR USE ON
80			BALL	A03AS02	3	4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL RB LO	0	0		WELD CAPS AND BLIND FLANGES
90			BALL	A03AS07	6	8	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION SOFT RB GO	0	0	FS1	USE FLANGES, GASKETS & VALVES AS
100			BALL	A03AS08	6	8	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION METAL RB GO	0	0		REQUIRED TO MATCH EQUIPMENT OR VESSEL
110			BALL	B03AS01	0.5	2	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB LO	0	0		NOZZLES.
120		VA5	BALL	B03AS02	3	3	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB LO	0	0	SB6	USE JACKSCREWS FOR PADDLE
130			BALL	B03AS03	4	4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING SOFT FB GO	0	0		BLANKS/SPACERS AND FLANGE SPREADERS
140		VA5	BALL	B03AS04	4	4	BALL 300# RF SS A182 F316 316SS TRIM FLOATING METAL FB GO	0	0		FOR SPECTACLE BLINDS.
150		VA5	BALL	B03AS07	10	24	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION SOFT FB GO	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL
160	U04	VA5	BALL	B03AS08	6	24	BALL 300# RF SS A182 F316 316SS TRIM TRUNNION METAL FB GO	0	0		CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR
170	U04	VA5	BALL								DBB PRESSURE CONNECTIONS WITH
180											CHEMICAL SEALS.
190										UK9	DO NOT USE FOR ORIFICE FLANGE TAP
200											ASSEMBLIES.
210							***** BUTTERFLY VALVES *****			VA5	USE THESE VALVES ONLY WHERE SPECIFIED
210											ON PIPING AND INSTRUMENT DIAGRAMS
220			BUTTERFLY	T03AS61	10	24	BUTTERFLY 300# RF SS A351 CF8M 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION
230											PURPOSE.
230							***** GATE VALVES *****			VAW	VALVE ONLY TO BE USED IN ACCORDANCE
240											WITH DOCUMENT L-ST-2055 - VENT DRAIN AND
240		ZA8	GATE	G03AS48	0.5	1.5	GATE 300# RF SS A182 F316 TRIM 16 WEDGE HW	0	0		INSTRUMENT CONNECTION DETAILS.
250	U04	ZA8	GATE	G03AS41	2	6	GATE 300# RF SS A351 CF8M TRIM 16 WEDGE HW	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO
260	U04	ZA8	GATE	G03AS41	8	8	GATE 300# RF SS A351 CF8M TRIM 16 WEDGE HW	0	0		MAXIMUM TEMPERATURE OF 200°C.
270										VC1	SPRING PISTON CHECK VALVES ARE PREFERRED
270							***** GLOBE VALVES *****				IN THE HORIZONTAL POSITION, BUT MAY BE
280											USED IN THE VERTICAL POSITION WITH
280			GLOBE	L03AS48	0.75	1.5	GLOBE 300# RF SS A182 F316 TRIM 16 OS&Y HW	0	0	ZA8	ENGINEERING APPROVAL.
290			GLOBE	L03AS41	2	6	GLOBE 300# RF SS A351 CF8M TRIM 16 OS&Y HW	0	0		*****TO BE USED IN STEAM AND CONDENSATE
300			GLOBE	L03AS42	8	10	GLOBE 300# RF SS A351 CF8M TRIM 16 OS&Y GO	0	0		SERVICE ONLY*****
310											
310							***** CHECK VALVES *****				
320											
320	U04	VC1	CHECK	C03AS50	0.5	1.5	CHECK PISTON 300# RF SS A182 F316 TRIM 16	0	0		
330	U04		CHECK	C03AS45	2	10	CHECK DUAL WAFER LUG 300# RF SS A351 CF8M TRIM 16 THROU BOLTED	0	0		
340	U04		CHECK	C03AS52	12	24	CHECK DUAL DBL FLG 300# RF SS A351 CF8M TRIM 16	0	0		
350											
350							*****PIPE*****				
360											
360	U04		PIPE	5365140	0.5	1.5	PIPE SCH 80S SMLS SS A312 TP316/316L PE	0	0		
370	U04	UK9	PIPE	5365202	0.75	0.75	PIPE SCH XXS SMLS SS A312 316/316L SS PE	0	0		
380			PIPE	5365136	2	12	PIPE SCH 40S SMLS SS A312 TP316/316L BE	0	0		
390			PIPE	5365137	14	24	PIPE SCH 40 SMLS SS A312 TP316/316L BE	0	0		
400		814	PIPE	5365207	0.5	0.5	PIPE SCH 160 SMLS SS A312 TP316/316L PE	0	0		
410											
410							***** FITTINGS *****				
420											
420	U04		ELBOW	5449033	0.5	1.5	ELL 90 DEG LR SCH 80S SS A403 WP316/316L-S	0	0		
430	U04	UK9	ELBOW	5449572	0.5	0.75	ELL 90 DEG LR SCH XXS SS A403 WP316/316L-S	0	0		
440			ELBOW	5449138	2	12	ELL 90 DEG LR SCH 40S SS A403 WP316/316L-S	0	0		
450			ELBOW	5449032	14	24	ELL 90 DEG LR SCH 40 SS A403 WP316/316L-S	0	0		
460	U04		ELBOW	5449146	0.5	1.5	ELL 45 DEG LR SCH 80S SS A403 WP316/316L-S	0	0		
470			ELBOW	5449144	2	12	ELL 45 SCH 40S SS A403 WP316/316L-S	0	0		
480	U04	UK9	ELBOW	5449143	0.5	0.75	ELL 45 DEG LR XXS SS A403 WP316/316L-S	0	0		
490			ELBOW	5449038	14	24	ELL 45 SCH 40 SS A403 WP316/316L-S	0	0		
500											
500	U04		TEE	5449156	0.5	1.5	TEE SCH 80S SS A403 WP316/316L-S	0	0		
510	U04	UK9	TEE	5449191	0.5	0.75	TEE SCH XXS SS A403 WP316/316L-S	0	0		
520			TEE	5449147	2	12	TEE SCH 40S SS A403 WP316/316L-S	0	0		
530			TEE	5449045	14	24	TEE SCH 40 SS A403 WP316/316L-S	0	0		
540			TEE RED	5448905	0.75	1	TEE RED SCH 80S SS A403 WP316/316L-S	1	1.5		
550			TEE RED	5449250	1	1.5	TEE RED SCH 40S/80S SS A403 WP316/316L-S	2	3		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		All SIZES - BUTT WELD SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 2.2 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 316/316L SS DUAL GRADE FLANGE RATING: 300 RF TEMPERATURE RANGE: -40 / 350 °C	12/16/2024	BY OZAC	U04	Page 1 of 3	300P22

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560			TEE RED	5449184	2	10	TEE RED SCH 40S SS A403 WP316/316L-S	3	12	
570			TEE RED	5448897	6	12	TEE RED SCH 40/40S SS A403 WP316/316L-S	14	24	
580			TEE RED	5449046	14	20	TEE RED SCH 40 SS A403 WP316/316L-S	16	24	
590										
590	U04		CAP	5449049	0.5	1.5	CAP SCH 80S SS A403 WP316/316L-S	0	0	
600			CAP	5449148	2	12	CAP SCH 40S SS A403 WP316/316L-S	0	0	
610			CAP	5449047	14	24	CAP SCH 40 SS A403 WP316/316L-S	0	0	
620										
620			RED CONC	5449158	0.5	0.75	REDUCER CONC SCH 80S SS A403 WP316/316L-S	1	1.5	
630			RED CONC	5549110	0.75	1.5	REDUCER CONC SCH 40S/80S SS A403 WP316/316L-S	2	4	
640			RED CONC	5449149	2	10	REDUCER CONC SCH 40S SS A403 WP316/316L-S	3	12	
650			RED CONC	5448902	6	12	REDUCER CONC SCH 40/40S SS A403 WP316/316L-S	14	20	
660			RED CONC	5448903	14	20	REDUCER CONC SCH 40 SS A403 WP316/316L-S	16	24	
670			RED ECC	5449159	0.5	0.75	REDUCER ECC SCH 80S SS A403 WP316/316L-S	1	1.5	
680			RED ECC	5549310	0.75	1.5	REDUCER ECC SCH 40S/80S SS A403 WP316/316L-S	2	4	
690			RED ECC	5449150	2	10	REDUCER ECC SCH 40S SS A403 WP316/316L-S	3	12	
700			RED ECC	5449154	6	12	REDUCER ECC SCH 40/40S SS A403 WP316/316L-S	14	20	
710			RED ECC	5449155	14	20	REDUCER ECC SCH 40 SS A403 WP316/316L-S	16	24	
720										
720	U04	FF5	FLANGE	5504515	0.75	1.5	NIPOFLANGE FLAT BOTTOM 300# RF SCH 160 LTCS A350 LF2 CL.1	0	0	
730			OLETS	5504411	0.75	1.5	NIPOFLANGE 300# RF SCH 40S/80S SS A182 F316/316L	2	12	
740			OLETS	5504421	0.75	1.5	NIPOFLANGE 300# RF SCH 40/80S SS A182 F316/316L	14	24	
750			OLETS	5504032	2	2	NIPOFLANGE 300# RF SCH 40S/40S SS A182 F316/316L	2	12	
760			OLETS	5504033	2	2	NIPOFLANGE 300# RF SCH 40/40S SS A182 F316/316L	14	24	
770		UJ2	OLETS	5504132	1.5	1.5	NIPOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4	12	
780		UJ2	OLETS	5504142	1.5	1.5	NIPOFLANGE 300# RF SCH 40/160 SS A182 F316/316L	14	24	
790	U04	FF5	OLETS	5504520	0.75	1.5	NIPOFLANGE FLAT BOTTOM 300# RF SCH 80S SS A182 F316/316L	0	0	
800										
800		UJ2	OLETS	5510354	0.5	1.5	WELDOLET SCH 40S/80S SS A182 F316/316L	2	12	
810			OLETS	5510364	0.75	1.5	WELDOLET SCH 40/80S SS A182 F316/316L	14	24	
820			OLETS	5509631	2	4	WELDOLET SCH 40S/40S SS A182 F316/316L	6	12	
830			OLETS	5509636	2	8	WELDOLET SCH 40/40S SS A182 F316/316L	14	24	
840	U04		OLETS	5504375	1	1.5	WELDOLET FLAT BOTTOM SCH 80S SS A182 F316/316L	0	0	
850										
850		UJ2	ELBOFLG	5516152	1.5	1.5	ELBOFLANGE 300# RF SCH 40S/160 SS A182 F316/316L	4	12	
860		UJ2	ELBOFLG	5516158	1.5	1.5	ELBOFLANGE 300# RF SCH 40/160 SS A182 F316/316L	14	24	
870										
870							***** FLANGES *****			
880										
880			FLANGE	5851203	0.5	2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0	
890										
890			FLANGE	5582991	0.5	1.5	FLG WN 300# RF SS A182 F316/316L SCH 80S BORE	0	0	
900			FLANGE	5582995	2	12	FLG WN 300# RF SS A182 F316/316L SCH 40S BORE	0	0	
910			FLANGE	5582997	14	24	FLG WN 300# RF SS A182 F316/316L SCH 40 BORE	0	0	
920	U04	FS1	FLANGE	5583535	2	6	FLG WN 600# RF SS A182 F316/316L SCH 40S BORE	0	0	
930	U04	FS1	FLANGE	5583021	3	6	FLG WN 900# RF SS A182 F316/316L SCH 40S BORE	0	0	
940	U04	FS1	FLANGE	5583970	2	2	FLG WN 1500# RF SS A182 F316/316L SCH 40S BORE	0	0	
950	U04	FS1	FLANGE	5583979	1.5	1.5	FLG WN 1500# RF SS A182 F316/316L SCH 80S BORE	0	0	
960	U04	FS1	FLANGE	5582998	0.5	1.5	FLG WN 600# RF SS A182 F316/316L SCH 80S BORE	0	0	
970		FA8	TEE RED	5600656	0.5	0.5	FLG SCRD (FNPT) REDUCING 300# RF SS A182 F316/316L	0.75	2	
980			BLIND	5616529	0.5	24	FLG BLIND 300# RF SS A182 F316/316L	0	0	
990										
990							***** ORIFICE FLANGES *****			
1000										
1000			ORIFICE FLG	5640598	2	12	WN ORIF SET 300# RF SS A182 F316/316L SCH 40S BORE	0	0	
1010			ORIFICE FLG	5640599	14	24	WN ORIF SET 300# RF SS A182 F316/316L SCH 40 BORE	0	0	
1020										
1020							***** PIPE NIPPLES *****			
1030										
1030		814		0			NIPPLE PBE CUT FROM PIPE			
1040										
1040							***** GASKETS *****			
1050										
1050			GASKET	5673033	0.5	24	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1060	U04		GASKET	5673017	3	3	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.5	0	0	
1070	U04		GASKET	5673103	0.5	6	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1080	U04		GASKET	5672251	1.5	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1090	U04		GASKET	5673117	3	6	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1100										
1100		B1					***** BOLTING *****			
1110										
1110	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1120										

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1120		SB6	BOLTING	5675675	12	24	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1130							***** BLINDS & SPACERS *****			
1140			SPEC BLIND	5750422	0.75	10	SPECTACLE BLIND 300# RF SS A240 GR 316	0	0	
1150			PADDLE	5750522	12	24	PADDLE BLANK 300# RF SS A240 GR 316	0	0	
1160			PADDLE	5750622	12	24	PADDLE SPACER 300# RF SS A240 GR 316	0	0	
1170										
1180							*****MISCELLANEOUS*****			
1190										
1190	U04		FLG GUARD	5558882	0.75	10	SAFETY SHIELD PTFE SUITABLE FOR 300# FLANGE	0	0	

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